Lecture plan · for the lecturer

Backpropagation

AI · University · Lecture 2

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Draw a network as a graph.
- Backpropagate by hand.
- Explain vanishing gradients.

Activity: Three nodes, real numbers

1. Warm-up (5 min): Ask students to explain in one sentence how a network decides which weights to change.
2. Model (15 min): Work forward and backward through a three-node graph with numbers on the board.
3. Practise (20 min): Pairs backpropagate through one sigmoid neuron with squared loss.
4. Check (15 min): Compare each hand gradient with (L(w + h) - L(w - h)) / 2h in code.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Where did your hand gradient differ from the check?
- Why do residual connections help?
- What should students derive, and what can they import?

Exit ticket

- Backpropagate through $y = (2w)^2$ at $w = 1$.

Try it in your next lecture
Draw a three-node graph with real numbers; pairs compute every gradient by hand, then check against the framework's answer.

Seminar sheet · for students

Backpropagation

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Name _____ Group _____ Date _____

1 **Why does the gradient shrink back through many sigmoid layers?**

◯ The learning rate is too small ◯ Each local derivative is ≤ 0.25 and they multiply

◯ Early layers have fewer weights

2 **Backpropagate through $L = (wx - y)^2$ with $w = 2$, $x = 3$, $y = 4$.**

The residual $wx - y$ = _____ , and $\partial L/\partial w = 2(wx - y)x$ = _____ .

Word bank: 4 · 12 · 2 · 6

3 **Which help gradients survive many layers? Choose all that apply.**

☐ Careful initialisation ☐ Adding more sigmoid layers ☐ Residual connections

☐ ReLU activations

4 **Computing all gradients by backpropagation costs roughly as much as one forward pass, not one pass per weight.**

◯ True ◯ False

5 **Match each student error to the misconception behind it.**

Adds local gradients instead of multiplying	• • • •	Ignores fan-out in the graph
Forgets to sum gradients where a value is used twice	• • • •	Confuses gradients with the optimiser
Thinks backprop updates the weights	• • • •	Misreads the chain rule

Answers · for the lecturer

Backpropagation

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1 **Each local derivative is ≤ 0.25 and they multiply**

A product of many numbers below 0.25 quickly approaches zero.

2 **The residual $wx - y = \underline{2}$, and $\partial L/\partial w = 2(wx - y)x = \underline{12}$.**

$wx - y = 6 - 4 = 2$; $\partial L/\partial w = 2 \times 2 \times 3 = 12$.

3 **ReLU activations; Residual connections; Careful initialisation**

ReLU has derivative 1 where active, residual paths add an identity route, and good initialisation keeps signals near unit scale.

4 **True**

Reverse-mode differentiation reuses the cached forward values, so its cost is a small constant times the forward pass.

5 **Adds local gradients instead of multiplying — Misreads the chain rule**
Forgets to sum gradients where a value is used twice — Ignores fan-out in the graph
Thinks backprop updates the weights — Confuses gradients with the optimiser

Local derivatives multiply along a path (the chain rule); where a value feeds two nodes, the gradients from both paths add (fan-out); and backprop only computes gradients, while the optimiser changes the weights.

For you to think about

How do you currently teach backpropagation? Where could a concrete three-node example replace notation?